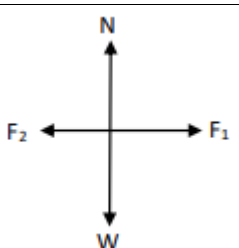


ANSWER SCHEME DAQ SES DP014 2023/2024_SET A

NO	ANSWER SCHEME	MARK												
1	<table border="1"> <thead> <tr> <th>Force</th><th>x-component</th><th>y-component</th></tr> </thead> <tbody> <tr> <td>F_1</td><td>$60 \cos 50^\circ$</td><td>$60 \sin 50^\circ$</td></tr> <tr> <td>F_2</td><td>$-200 \cos 40^\circ$</td><td>$200 \sin 40^\circ$</td></tr> <tr> <td></td><td>-114.64</td><td>174.52</td></tr> </tbody> </table> $F = \sqrt{F_x^2 + F_y^2}$ $F = \sqrt{(-114.64)^2 + (174.52)^2}$ $F = 208.81N$ $\tan \theta = \frac{174.52}{-114.64}$ $\theta = -56.7^\circ$ Above negative x-axis	Force	x-component	y-component	F_1	$60 \cos 50^\circ$	$60 \sin 50^\circ$	F_2	$-200 \cos 40^\circ$	$200 \sin 40^\circ$		-114.64	174.52	J1 J1 J1 G1 JU1 G1 JU1 J1
Force	x-component	y-component												
F_1	$60 \cos 50^\circ$	$60 \sin 50^\circ$												
F_2	$-200 \cos 40^\circ$	$200 \sin 40^\circ$												
	-114.64	174.52												
	TOTAL	8												
2(a) (i)		SHAPE – 1 LABEL AXIS WITH CORRECT UNIT – 1 LABEL VALUE – 1												
ii	$a = \frac{\Delta v}{\Delta t}$ $0.5 = \frac{\Delta v}{60}$ $v \text{ (m/s)}$ $v = 30 \text{ m s}^{-1}$ $a = \frac{\Delta v}{\Delta t}$ $a = \frac{0 - 30}{1800 - 1440}$ $a = -0.083 \text{ m s}^{-2}$ =0.0217 ms-2	G1 JU1 G1 JU1												

	s = under the graph $s = \frac{1}{2}(30)(60)$ $s = 900 \text{ m}$ $s = (30)(1380) \text{ (420-60)}$ $s = 41400 \text{ m } 10800 \text{ m}$ $s = \frac{1}{2}(30)(360) \text{ (1800-420)}$ $s = 5400 \text{ m } 20700 \text{ m}$ Distance travelled = 900 + 41400 + 5400 = 47700 m = 900 + 10800 + 20700 = 32400 m OR $s = \frac{1}{2}(a + b)c$ $s = \frac{1}{2}(\text{recalculate } 180 + 3600)(30)$ $s = 47700 \text{ m}$	K1 G1 JU1 K1 G1 JU1
iii	$s = ut + \frac{1}{2}at^2$ $800 = 10t + \frac{1}{2}(3)t^2$ $t = 20.0 \text{ s}$ Yes, he able to get to his destination	G1 JU1 J1
	TOTAL	13
3a(i)	$J = m(v - u)$ $J = (0.060)(-8 - 10)$ $J = -1.08 \text{ N s}$	G1 JU1
(ii)	$J = F \Delta t$ $F = \frac{1.08}{(3.0 \times 10^{-3})}$ $F = -360.00 \text{ N}$	G1 JU1
b)	$\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$ $mu_1 + mu_2 = mv_{1x} + mv_{2x}$ $5(2.0) + 5(0) = 5(1.7 \cos 30^\circ) + 5(v_{2x})$ $v_{2x} = 0.527 \text{ m/s}$	K1 G1

	$0 = mv_{1y} + mv_{2y}$ $0 = 5(1.7 \sin 30^\circ) + 5(v_{2y})$ $v_{2y} = -0.85 \text{ m s}^{-1}$ $v = \sqrt{(0.527)^2 + (-0.85)^2}$ $v_2 = 1.00 \text{ m/s}$	G1 G1 JU1
	Total	9
4 (a)	 Correct arrow and label force 4 forces – 2m 3 forces – 1m 2 forces – 0 m	D2
(ii)	$\Sigma F_x = ma$ $F_1 - F_2 = ma$ $10 - 8 = 5a$ $a = 0.4 \text{ ms}^{-2}$ Direction: to the right.	K1 G1 JU1 J1
(b)	$\Sigma F_y = 0$	K1
(i)	$N - W_y = 0$ $N = mg \cos 30^\circ$ $N = (10)(9.81) \cos 30^\circ$ $N = 84.96 \text{ N}$	G1 JU1
(ii)	$f = \mu N$ $f = (0.1)(84.96)$ $f = 8.496 \text{ N}$	GJU1
(iii)	$\Sigma F_x = W_x - f$ $\Sigma F_x = (10)(9.81) \sin 30^\circ - 8.496$ $\Sigma F_x = 40.554 \text{ N}$	G1 JU1
	Total	12 Marks

5 (a) i.	$a_c = r\omega^2$ $6.5 = (1.20) \omega^2$ $\omega = 2.33 \text{ rad s}^{-1}$ $\omega = (2.33 \text{ rad s}^{-1})\left(\frac{1 \text{ rev}}{2\pi \text{ rad}}\right)\left(\frac{60 \text{ s}}{1 \text{ min}}\right)$ $\omega = 22.25 \text{ rev min}^{-1}$	G 1 J 1 J U 1
(a) ii)	$\Sigma F_x = F_c = T = ma_c$ $T = (0.18)(6.5)$ $T = 1.17 \text{ N}$	K 1 G J U 1
5 (b) i)	$a_c = v^2/r$ $r = 16^2/15$ $r = 17.07 \text{ m}$ $a_c = v^2/r$ $a_c = 8^2/17.07$ $a_c = 3.75 \text{ m s}^{-2}$	 J 1 G J U 1
(b) ii)	$f_s = F_c = ma_c$ $f_s = (2000)(3.75)$ $f_s = 7500 \text{ N}$	G 1 J U 1
	TOTAL	9 M
6 (a)(i)	$\omega_0 = 600 \text{ rpm} \left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) \left(\frac{1 \text{ minute}}{60 \text{ s}}\right) = 62.83 \text{ rad s}^{-1}$ $\omega_f = 300 \text{ rpm} \left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) \left(\frac{1 \text{ minute}}{60 \text{ s}}\right) = 31.42 \text{ rad s}^{-1}$ $\omega_f = \omega_0 + \alpha t$ $\alpha = \frac{\omega_f - \omega_0}{t} = \frac{31.42 - 62.83}{6}$ $\alpha = -5.235 \text{ rad s}^{-2}$	J 1 J 1 G 1 J U 1
(a) (ii)	$\omega_f^2 = \omega_0^2 + 2\alpha\theta$ $\theta = \frac{\omega_f^2 - \omega_0^2}{2\alpha} = \frac{31.42^2 - 62.83^2}{2(-5.235)}$ $\theta = 282.75 \text{ rad}$ No of revolutions $= 282.75 \text{ rad} \left(\frac{1 \text{ rev}}{2\pi \text{ rad}}\right) = 44.99 \text{ revs}$	 G 1 J 1 J U 1

	$\frac{dQ}{dt} = -kA \frac{dT}{x}$ $\frac{Q}{120} = -(385)(4 \times 10^{-4})(-67.5)$ $\therefore Q = 1247.4 \text{ J}$	G1 JU1
7 (b)	Given : M of nitrogen gas = $0.028 \text{ kg mol}^{-1}$; M of the oxygen gas = $0.032 \text{ kg mol}^{-1}$ $T = 45^{\circ} \text{ C} = 273.15 + 45 = 318.15 \text{ K}$ $v_{rms} = \sqrt{\frac{3RT}{M}}$ for nitrogen gas , $v_{rms} = \sqrt{\frac{(3)(8.31)(318.15)}{0.028}}$ $\therefore v_{rms} = 5.32 \times 10^2 \text{ m s}^{-1}$ for oxygen gas , $v_{rms} = \sqrt{\frac{(3)(8.31)(318.15)}{0.032}}$ $\therefore v_{rms} = 4.98 \times 10^2 \text{ m s}^{-1}$	G1 JU1 G1 JU1
7 (c)	Given : $P = 1 \text{ atm} = P = 1.013 \times 10^5 \text{ Pa}$ $V_i = 0.03 \text{ m}^3$ $V_f = 0.07 \text{ m}^3$ $Q = 400 \text{ J}$ i) $W = P\Delta V$ $W = (1.013 \times 10^5)(0.07 - 0.03)$ $\therefore W = 4125.2 \text{ J}$ ii) $\Delta U = Q - W$ $\Delta U = 400 - 4125.2$ $\therefore \Delta U = -3725.2 \text{ J}$	G1 JU1 G1 JU1
7 (d)	i) $W_{AB} = 0 \text{ J}$	JU1

	$W = P\Delta V$ $W_{BC} = (5 \times 1.0313 \times 10^5)(0.08 - 0.04) = 20626 J$ • <i>work done by system</i> (W = +ve) ii) Given : Q = +3600 J $\Delta U_{BC} = 3600 - 20626$ $\therefore \Delta U_{BC} = -17026 J$	GJU1 J1 K1 G1 JU1
	TOTAL MARKS	18
	GRAND TOTAL	80